

The difference and the sum of two cubes

Two factorisations that a sum of squares never had — and the trinomial that will not factorise further.

LESSON 104–105

2 × 40 MIN

MATEMATIKA 7, §47

BEYOND STAGE 8

LESSON CLOCK

15'

20'

25'

20'

5'

The two formulae	15 min
Why the middle terms vanish	20 min
Factorising	25 min
Where they are used	20 min
Homework	5 min

LEARNING OBJECTIVES

- Verify $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ by expanding.
- Factorise sums and differences of cubes.
- Distinguish the incomplete square $a^2 + ab + b^2$ from $(a + b)^2$.
- Use the formulae in numerical work and in cancelling fractions.

TEXTBOOK

National	pp. 304–310
Cambridge	Stage 8 extension
Workbook	Extension sheet 2B

01

Explanation

The two formulae

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

FORMULA	SIGN IN THE BRACKET OF TWO	SIGN OF THE MIDDLE TERM
$a^3 - b^3$	–	$+ab$
$a^3 + b^3$	+	$-ab$

THE SIGNS ARE ALWAYS OPPOSITE

Whatever sign stands between a and b in the short bracket, the middle term of the long bracket takes the other one. Getting these two the same way round is the only real difficulty in the topic.

Why the middle terms vanish

Expand $(a - b)(a^2 + ab + b^2)$ in full and watch four of the six terms cancel.

PRODUCT	TERMS
$a \cdot (a^2 + ab + b^2)$	$a^3 + a^2b + ab^2$
$-b \cdot (a^2 + ab + b^2)$	$-a^2b - ab^2 - b^3$
sum	$a^3 - b^3$

$A^2 + AB + B^2$ IS NOT $(A + B)^2$

The square has $2ab$ in the middle; this trinomial has just ab . It is called the **incomplete square** for exactly that reason, and it does not factorise any further.

Factorising

EXPRESSION	A	B	FACTORISED
$x^3 - 8$	x	2	$(x - 2)(x^2 + 2x + 4)$
$x^3 + 27$	x	3	$(x + 3)(x^2 - 3x + 9)$
$8a^3 - 1$	$2a$	1	$(2a - 1)(4a^2 + 2a + 1)$
$27x^3 + 64y^3$	$3x$	$4y$	$(3x + 4y)(9x^2 - 12xy + 16y^2)$

A SUM OF CUBES DOES FACTORISE

$a^2 + b^2$ was a dead end, but $a^3 + b^3$ is not — the odd power makes all the difference. That is worth saying out loud, because the two look alike on the page.

Where they are used

TASK	WORKING	ANSWER
$\frac{x^3 - 8}{x - 2}$	$\frac{(x - 2)(x^2 + 2x + 4)}{x - 2}$	$x^2 + 2x + 4$
$\frac{x^3 + 27}{x + 3}$	cancel $x + 3$	$x^2 - 3x + 9$
$10^3 - 1$	$9 \cdot (100 + 10 + 1)$	999

WHY $N^3 - 1$ IS ALWAYS DIVISIBLE BY $N - 1$

The factorisation shows it directly: one factor is $n - 1$. The same argument explains why 999, 9999 and every repunit-like number of that shape has an obvious divisor.

02 Worked examples

EXAMPLE 1 Factorise $8a^3 - 1$.

- ① $a = 2a$ and $b = 1$ — the cube roots.
- ② Short bracket: $(2a - 1)$.
- ③ Long bracket: $(2a)^2 + 2a \cdot 1 + 1^2 = 4a^2 + 2a + 1$.
Middle sign is the opposite.
- ④ $= (2a - 1)(4a^2 + 2a + 1)$

ANSWER

$$(2a - 1)(4a^2 + 2a + 1)$$

EXAMPLE 2 Factorise $27x^3 + 64y^3$.

① $a = 3x, b = 4y.$

② Short bracket: $(3x + 4y).$

③ Long bracket: $9x^2 - 12xy + 16y^2.$
Plus outside, minus inside.

④ $= (3x + 4y)(9x^2 - 12xy + 16y^2)$

ANSWER

$$(3x + 4y)(9x^2 - 12xy + 16y^2)$$

EXAMPLE 3 Simplify $\frac{x^3 - 8}{x - 2}$.

① $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$

② The factor $x - 2$ cancels.

③ $= x^2 + 2x + 4$
For $x \neq 2$.

ANSWER

$$x^2 + 2x + 4$$

03 Interactive model

Ask the class to expand $(a - b)(a^2 + ab + b^2)$ themselves before the formula is given; the six terms collapsing to two is more convincing than any statement of the rule.

A cube factorised, then cancelled

$$\frac{x^3 - 8}{x - 2}$$

① $a = x$

and $b = 2$

The
cube
roots
of
the
two
terms.

② $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$

Minus
outside,
plus
inside.

③ $x^2 + 2x + 4$

Cancel
 $x -$
2;
note
 $x \neq$
2.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Difference of two cubes	Kublar ayirmasi	Разность кубов
Sum of two cubes	Kublar yig'indisi	Сумма кубов
Incomplete square	To'liqsiz kvadrat	Неполный квадрат
Cube root	Kub ildiz	Кубический корень
Trinomial	Uch had	Трёхчлен
Opposite signs	Qarama-qarshi ishoralar	Противоположные знаки
To verify	Tekshirmoq	Проверить
Factorisation	Ko'paytuvchilarga ajratish	Разложение на множители

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a^3 - b^3$ equals:

$$(a - b)(a^2 + ab + b^2)$$

$$(a - b)^3$$

$$(a - b)(a^2 - ab + b^2)$$

$$(a - b)(a + b)^2$$

$a^3 + b^3$ equals:

$$(a + b)^3$$

$$(a + b)(a^2 - ab + b^2)$$

$$(a + b)(a^2 + ab + b^2)$$

it does not factorise

$x^3 - 8$ factorises to:

$$(x - 2)(x^2 + 2x + 4)$$

$$(x - 2)^3$$

$$(x - 2)(x^2 - 2x + 4)$$

$$(x - 8)(x^2 + 1)$$

$a^2 + ab + b^2$ is:

$(a + b)^2$

the incomplete square

$(a - b)^2$

a difference of squares

$x^3 + 27$ factorises to:

$(x + 3)(x^2 - 3x + 9)$

$(x + 3)^3$

$(x + 3)(x^2 + 3x + 9)$

it does not

Compared with $a^2 + b^2$, a sum of cubes:

also fails to factorise

does factorise

is a square

is negative

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x^3 - 8$

ANSWER $(x - 2)(x^2 + 2x + 4)$

2. $x^3 + 8$

ANSWER $(x + 2)(x^2 - 2x + 4)$

3. $x^3 - 1$

ANSWER $(x - 1)(x^2 + x + 1)$

4. $x^3 + 1$

ANSWER $(x + 1)(x^2 - x + 1)$

5. $x^3 + 27$

ANSWER $(x + 3)(x^2 - 3x + 9)$

6. $x^3 - 64$

ANSWER $(x - 4)(x^2 + 4x + 16)$

7. The name of $a^2 + ab + b^2$

ANSWER the incomplete square

Medium · the standard question

7 PROBLEMS

1. $8a^3 - 1$

ANSWER $(2a - 1)(4a^2 + 2a + 1)$

2. $27x^3 + 64y^3$

ANSWER $(3x + 4y)(9x^2 - 12xy + 16y^2)$

3. $125 - a^3$

ANSWER $(5 - a)(25 + 5a + a^2)$

4.
$$\frac{x^3 - 8}{x - 2}$$

ANSWER $x^2 + 2x + 4$

5.
$$\frac{x^3 + 27}{x + 3}$$

ANSWER $x^2 - 3x + 9$

6. $10^3 - 1$ factorised

ANSWER $9 \cdot 111 = 999$

7. Does $a^3 + b^3$ factorise?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. $64x^3 - 27y^3$

ANSWER $(4x - 3y)(16x^2 + 12xy + 9y^2)$

2. $2x^3 + 16$

ANSWER $2(x + 2)(x^2 - 2x + 4)$

3. $a^6 - b^6$ as a product of four factors

ANSWER $(a - b)(a + b)(a^2 + ab + b^2)(a^2 - ab + b^2)$

4. Why is $n^3 - 1$ always divisible by $n - 1$?

ANSWER It is one of the factors

5. $x^3 - \frac{1}{8}$

ANSWER $(x - \frac{1}{2})(x^2 + \frac{x}{2} + \frac{1}{4})$

6. If $a + b = 5$ and $ab = 6$, find $a^3 + b^3$ using the formula

ANSWER 35

7. Factorise $x^6 - 64$ as a difference of squares first

ANSWER $(x^3 - 8)(x^3 + 8)$

07 Homework**Homework — 5 tasks**

Write the short bracket first; its sign decides the middle sign of the long one.

1. Factorise $x^3 - 125$ and $x^3 + 64$.

2. Factorise $27a^3 - 8b^3$.

3. Simplify $\frac{x^3 + 1}{x + 1}$.

4. Factorise $5x^3 - 40$ completely.

5. Explain in one sentence the difference between $a^2 + ab + b^2$ and $(a + b)^2$.